

Immersive Behavioral Therapy for Phobia Treatment in Individuals with Intellectual Disabilities

Carlos Cortés*
ETSI Telecomunicación
Universidad Politécnica de Madrid
Ainhoa Fernández-Alcaide
Fundación Juan XXIII

Marta Goyena
ETSI Telecomunicación
Universidad Politécnica de Madrid
María Nava-Ruiz
Fundación Juan XXIII

Marta Orduna
Nokia XR Lab
Jesús Gutiérrez
ETSI Telecomunicación
Universidad Politécnica de Madrid

Matteo Dal Magro
ETSI Telecomunicación
Universidad Politécnica de Madrid
Pablo Pérez
Nokia XR Lab

Narciso García
ETSI Telecomunicación
Universidad Politécnica de Madrid

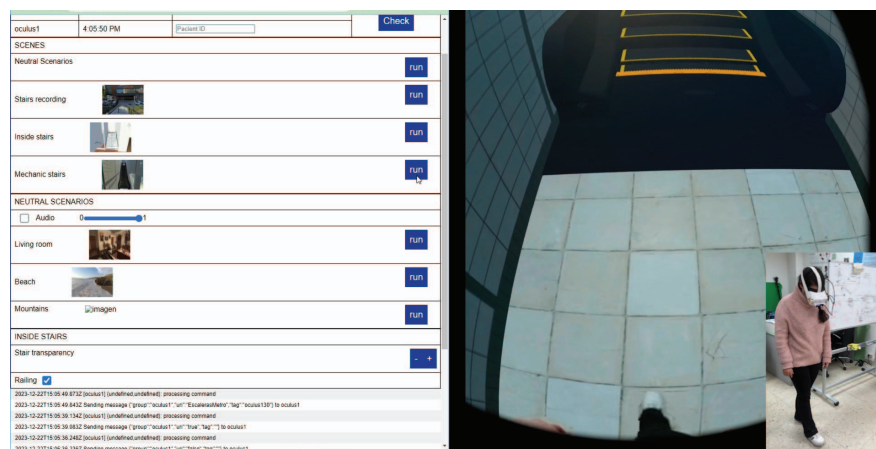


Figure 1: Picture of the therapist monitor: the left panel shows the control area and the right panel shows the user's viewport.

ABSTRACT

Adapting traditional behavioral therapy with eXtended Reality (XR) and physiological data allows phobia treatment for individuals with intellectual disabilities. Using systematic desensitization, our tool creates tailored virtual environments for stair-related phobias. Therapists guide the process, monitoring real-time data from the Head-Mounted Display and Empatica E4 wristband during the session. This approach promises a valid transition from classical to immersive therapies. A video demo can be seen in: <https://youtu.be/pCvxhyMpQ0>.

Index Terms: Human-centered computing—Accessibility—Accessibility systems and tools; Human-centered computing—Accessibility—Accessibility design and evaluation methods;

1 INTRODUCTION

XR technologies allow greater immersion of the patient in the environment and can provide the sense of being in remote places, which can enhance live therapies. In this sense, the use of these technologies can facilitate remote access to therapies (telecare), without having to travel to the corresponding center, as well as minimizing costs and resources by allowing the simulation of physical objects

and environments [3]. Likewise, these technologies allow the easy collection of parameters related to the patient's behavior from sensors integrated into the HMDs (e.g. position and navigation through the environment, head and eye exploration movements, etc.). It is also possible to obtain other parameters with non-intrusive devices and sensors that allow to measure physiological signals, such as heart rate or galvanic response of the skin. These parameters can be considered digital biomarkers, since, as has been demonstrated in several studies, can be useful in the identification of mental states and cognitive processes [4, 7, 8]. Obtaining digital biomarkers not only allows the study of their relationship with cognitive processes and the patient's state, but can also monitor the evolution of the patient with therapies. In addition, it also motivates the development of tools that allow the activation (automatically or by informing the therapist) of regulation of emotional or cognitive mechanisms [2].

Specifically, this work addresses the development of a practical tool for alleviating anxiety in individuals with intellectual disabilities who experience phobia in situations involving stairs. This practical tool is based on the systematic desensitization technique [1], used to help patients to manage the emotions produced by anxiety-provoking experiences and reduce the avoidance behaviors. This technique considers a training phase on relaxation, a practice in imagination and on the desensitization itself by facing the patient to anxiety-provoking stimuli. The inclusion of immersive technology has many advantages over the traditional therapy. The practice of imagination, which can be difficult for individual with intellectual disabilities, is replaced by viewing 360-degree videos. In addition, patients can face different stimuli in the same physical space without

* e-mail: carlos.cs@upm.es

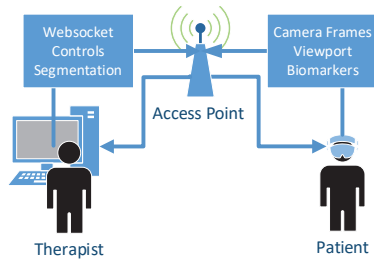


Figure 2: Connection diagram of tool components.

having to move to another location. In that context, a determining factor in obtaining the sensation of being physically in another place is the embodiment. Therefore, we have incorporated to the tool an advanced photorealistic body visualization system based on image segmentation through the use of neural networks [6]. While current systems allow the perception of the hands, ours allows the visualization of the whole body, including the feet, which is essential for the use case of this study (dealing with the fear of stairs). Moreover, immersive technologies also provide a more controlled scenario for the therapist to guide the intervention effectively taking into account the digital biomarkers, which is vital when dealing with users with intellectual disabilities.

The goal of this work is to implement the systematic desensitization technique in XR and integrate it with the control of biomarkers, applying immersive behavioral therapy sessions. The tool has evolved alongside the therapists who played a crucial role in the process of remodeling and technical design of the tool. Consequently, the main contribution of this research lies in the behavioral therapy tool adapted for individuals with intellectual disabilities and designed by and for therapists who regularly interact with them.

The therapist's monitor is shown in Fig 1. The application incorporates the following functionalities, which are detailed in the subsequent sections: 1) display of the user's viewport; 2) control by the therapist, being able to change the stimuli during the experience; 3) egocentric segmentation for natural body representation; and 4) visualization of biometric data.

2 TOOL FOR IMMERSIVE BEHAVIORAL THERAPY

To enhance behavioral therapy through the use of immersive technology, a tool has been developed that seamlessly integrates XR, physiological sensor data, and therapist monitoring. The hardware components include a Meta Quest 2 HMD with an USB stereo camera, a Windows PC running a neural network for body segmentation and hosting a web application for therapist control, and an Empatica E4 wristband for physiological data acquisition.

The software components that run on the HMD are an app developed using the Unity game engine to create immersive virtual scenarios and a Unity service to manage the camera frames, receive commands from the therapist and transmit them to the virtual environment. Two types of virtual scenarios have been implemented. The first type is based on the reproduction of 360-degree videos and the second one is based on 3D designs of spaces containing stairs. The software running from the PC includes a websocket server that manages the control commands between the therapist's web application and the Unity engine. It also includes an *adb* client called *scrpcy* for viewing the user's viewport wirelessly. Finally, to allow users to see their feet and body, an egocentric segmentation model, based on a neural network, runs on the control PC [6].

The relationship of the different components is illustrated in Fig. 2. From the patient or user side, the HMD transmits the frames of the USB camera, the viewport and the biomarkers to the therapist's PC through a wireless access point. From the therapist's point of view, the server performs session control through websockets. This server

is also responsible for segmenting the camera frames so that the user can perceive their body within the XR like in [5].

3 USE OF THE APPLICATION

Fig. 1 illustrates what the therapist visualizes while monitoring the session. The left panel shows the control area that allow the therapist to switch between the different scenarios for each phase of the systematic desensitization (i.e. relaxation scenes or neutral scenarios and stressful scenarios). For greater control over the session, therapists are also able to view the users' viewport through the right panel.

From the user's point of view, they will find themselves in a calibrated space, where they will first be shown pre-recorded 360-degree videos of relaxing environments, such as a beach or a living room. Afterwards, the therapist will use the control panel to vary between different anxiety-producing stimuli to generate desensitization. Furthermore, the therapist will see the information of the recorded biomarkers that will help him or her to have more information about the level of anxiety suffered by the user, thus, helping the therapist to make the decision of which scenes to select.

4 CONCLUSION AND FUTURE WORK

This work has shown the design and operation of a tool for the execution of behavioral therapies for the treatment of phobias based on systematic desensitization. Specifically, for treatment of stairs phobia, which is a everyday handicap in some individuals with intellectual disabilities. The tool has been developed with the collaboration of their therapists. Therefore, this tool aims at helping these patients to improve their lives through the opportunities that new technologies provide. As future work, we propose integrating real-time biomarker measurements into the main application. In addition, we will explore the use of the cameras that the new HMDs incorporate to avoid having to use extra hardware.

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REFERENCES

- [1] Y. Choy, A. J. Fyer, and J. D. Lipsitz. Treatment of specific phobia in adults. *Clinical psychology review*, 27(3):266–286, 2007.
- [2] D. Colombo, A. Díaz-García, J. Fernandez-Álvarez, and C. Botella. Virtual reality for the enhancement of emotion regulation. *Clinical Psychology & Psychotherapy*, 28(3):519–537, 2021.
- [3] C. Cortés, M. Rubio, J. Gutierrez, B. Sánchez, P. Pérez, and N. García. Evaluation of interaction methods in an extended reality training environment. In *2023 ACM International Workshop on Immersive Mixed and Virtual Environment Systems, MMVE '23*, p. 24–29, 2023.
- [4] G. Coughlan, A. Coutrot, M. Khondoker, A.-M. Minihi, H. Spiers, and M. Hornberger. Toward personalized cognitive diagnostics of at-genetic-risk alzheimer's disease. *Proceedings of the National Academy of Sciences*, 116(19):9285–9292, 2019.
- [5] D. Gonzalez Morin, E. Gonzalez-Sosa, P. Perez, and A. Villegas. Full body video-based self-avatars for mixed reality: from e2e system to user study. *Virtual Reality*, pp. 1–19, 2023.
- [6] E. Gonzalez-Sosa, A. Gajic, D. Gonzalez-Morin, G. Robledo, P. Perez, and A. Villegas. Real time egocentric segmentation for video-self avatar in mixed reality, 2022.
- [7] L. Itti. New eye-tracking techniques may revolutionize mental health screening. *Neuron*, 88(3):442–444, 2015.
- [8] K. Kyriakou et al. Detecting moments of stress from measurements of wearable physiological sensors. *Sensors*, 19(17):3805, 2019.